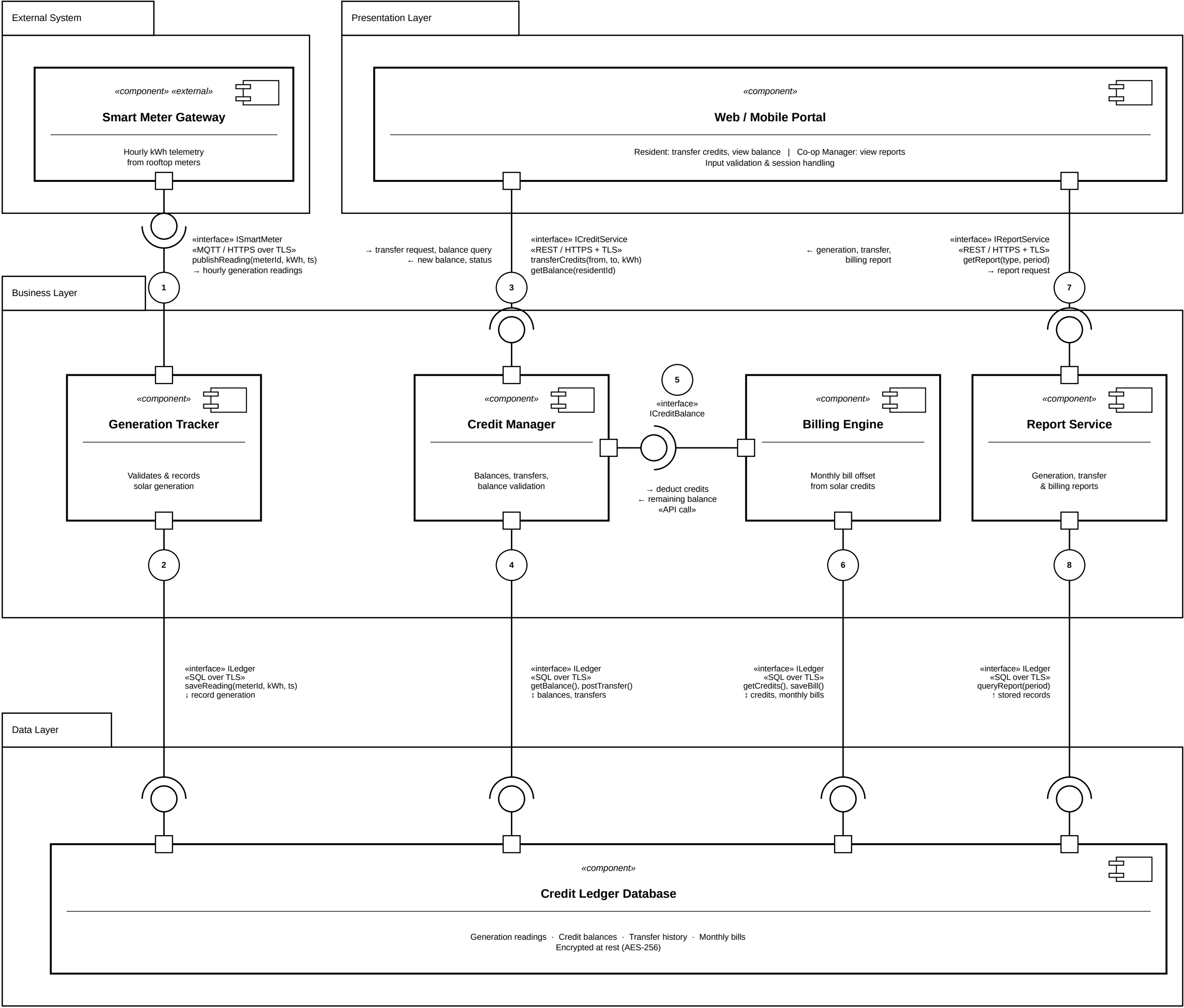


Community Solar Credit Allocation Manager — UML Component Diagram

Problem Statement #63 · Sustainability & Green Tech · Architectural style: Layered Architecture



Notation

«component»

Provided interface (ball)

Required interface (socket)

Port

Flow no.

Assembly

Interaction flows mapped to requirements

1 – 2 FR-001 Smart Meter → Generation Tracker (ISmartMeter) → readings stored in ledger (ILedger)

3 – 4 FR-002, FR-003 Portal → Credit Manager (ICreditService) → validates & updates balances in ledger

5 – 6 FR-004 Billing Engine → Credit Manager (ICreditBalance) → offset credits, save bill in ledger

7 – 8 FR-005 Portal → Report Service (IReportService) → report data read from ledger

NFR-001 / NFR-002: stateless Business Layer (replicable, 99.9% uptime); every link uses TLS, ledger encrypted at rest